SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Register Number	**192365013**
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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**

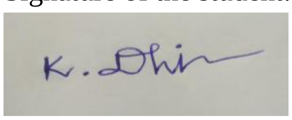
Question Count: 10

Code of Conduct

I **K.Dhilip/192365013** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

HDFS (Hadoop Distributed File System) is designed to store very large files across multiple computers in a Hadoop cluster. It divides files into blocks and stores replicated copies on different DataNodes.

A traditional file system usually stores files on a single server or storage device.

Key difference: HDFS provides distributed storage, fault tolerance, and scalability**,** whereas traditional file systems are mainly designed for local or centralized storage.

2. Why is replication important in distributed storage systems?

Replication means maintaining multiple copies of data on different nodes**.**

It is important because if one node fails, another copy can still be accessed. Therefore, replication provides:

- High data availability
- Fault tolerance
- Reliability
- Faster recovery

For example, HDFS can use a replication factor of 3**,** storing three copies of each block.

3. How does MapReduce divide work between map and reduce phases?

MapReduce divides a large dataset into smaller portions and processes them in parallel.

- **Map phase:** Reads input data and generates intermediate key-value pairs**.**
- **Shuffle and Sort:** Groups values with the same key.
- **Reduce phase:** Processes the grouped data and produces the final output.

Example: In word counting, the Mapper produces word → 1, and the Reducer adds the values for each word.

4. What role does YARN play in large-scale Hadoop processing?

YARN (Yet Another Resource Negotiator) is the resource management and job scheduling layer of Hadoop.

It:

- Allocates CPU and memory resources.
- Schedules applications.
- Manages containers.
- Monitors running applications.

- Allows multiple workloads to share the same cluster.

Thus, YARN ensures efficient and controlled use of cluster resources**.**

5. Differentiate NAS and SAN in an enterprise data environment.

NAS	**SAN**
Network Attached Storage	Storage Area Network
Provides File-Level Storage	Provides Block-Level storage
Users access shared files	Servers access storage blocks like local disks
Users protocols such as NFS/SMB	Commonly uses Fibre Channel or iSCSI
Easier and generally cheaper	More expensive and complex
Suitable file sharing	Suitable for databases and high-performance application

NAS provides shared files, while SAN provides high-performance block storage.

6. What is the purpose of ETL in a big data pipeline?

ETL means Extract, Transform, Load.

- **Extract:** Collect data from different sources.
- **Transform:** Clean, validate, filter, and convert the data.
- **Load:** Store the processed data in the target system.

ETL ensures that data is clean, consistent, and ready for analysis**.**

7. How does Apache NiFi help in data integration workflows?

Apache NiFi is a data-flow management tool used to collect, move, transform, and route data between different systems.

It provides:

- Data ingestion from multiple sources.
- Data routing and transformation.
- Secure data transfer.
- Error handling and retries.
- Data provenance and monitoring.

Therefore, NiFi helps organizations build automated and reliable data integration pipelines.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Major failure points include:

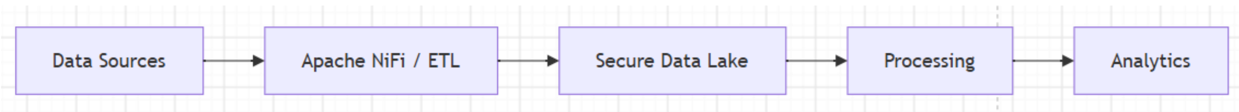
1. **DataNode failure** → Use data replication and automatic re-replication.

2. **NameNode failure** → Use NameNode High Availability (Active/Standby).
3. **ResourceManager failure** → Use YARN ResourceManager HA.
4. **Disk failure** → Replace failed disks and recover data from replicas.
5. **Network failure** → Use redundant network connections and proper monitoring.

The main principle is to avoid single points of failure and provide automatic recovery.

9. How would you design a secure data ingestion pipeline for a large organization?

Architecture:



Security can be provided by:

- TLS/SSL for data in transit.
- Kerberos for authentication.
- Role-Based Access Control (RBAC) for authorization.
- Encryption for sensitive data at rest.
- Data validation and sanitization.
- Audit logs and monitoring.
- Backup and replication for recovery.

This ensures secure, controlled, reliable, and traceable data movement.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Hadoop works like a large team of workers. Instead of one person processing a huge amount of data, Hadoop divides the work among many computers and processes it at the same time. When the amount of data grows, we can add more computers. If one computer fails, another can continue the work using a backup copy.

Therefore, Hadoop provides scalability, reliability, parallel processing, and cost-efficient data management.